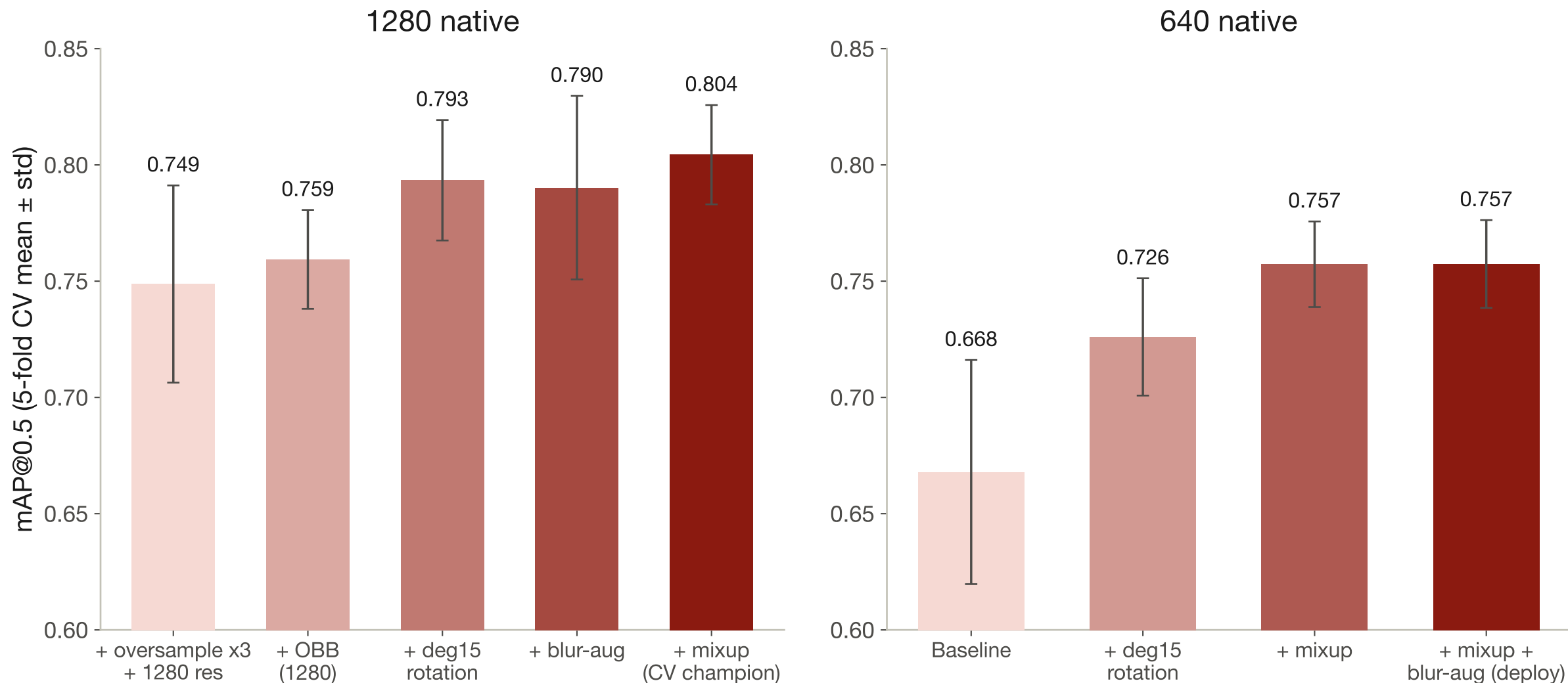


Training strategies, worst to best (YOLOv11n)



974-image eduardo pool, group-aware 5-fold CV means. Left = 1280-px native training; right = 640-px native training (current deployment resolution). Both ladders end the same way: rotation/scale augmentation, then mixup — at 640 mixup also composes with blur-aug for robustness at no clean-accuracy cost.